

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A Comparative Analysis

Code of Conduct:

*I **KESHAVARAM L/192511253** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

L. Keshav

Annexure A
Comparative Analysis

1. Apply **SSL/TLS and IPsec protocols** to a secure communication scenario and determine which provides better security for web-based and network-layer protection.
2. Design a secure email system by comparing **end-to-end encryption and gateway-based encryption**, and justify the most suitable approach for enterprise use.
3. Apply **HMAC and Digital Signature techniques** to a data transmission scenario and analyze which method provides better authentication and integrity.
4. Create a **spam detection system** by comparing supervised learning and unsupervised learning approaches, and evaluate their effectiveness in real-world email filtering.
5. Design a secure communication framework by comparing **AES and RSA encryption techniques**, and determine their roles in achieving confidentiality and efficiency.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. SSL/TLS vs IPsec for Secure Communication

Scenario

Consider an organization where employees access a web-based banking application through the Internet. The organization needs to protect both web communication and network-layer communication.

Comparison

Feature	SSL/TLS	IPsec
Security layer	Transport/Application layer	Network layer
Main purpose	Secures web and application communication	Secures IP packets
Common use	HTTPS, secure web applications	VPNs, site-to-site networks
Encryption	Yes	Yes
Authentication	Certificates, PSK, etc.	Certificates, PSK, etc.
Integrity	MAC/AEAD mechanisms	Authentication headers/AEAD
Scope	Specific application connection	Entire IP communication
Deployment	Relatively easy for web services	More complex
Example	HTTPS	IPsec VPN
Best suited for Web-based protection		Network-wide protection

Analysis

SSL/TLS is more suitable when the primary requirement is protecting web applications. For example, HTTPS uses TLS to protect login credentials, transactions and other application data.

IPsec operates at the network layer and can protect communication between entire networks or hosts. It is therefore highly suitable for VPNs and secure communication between branch offices.

Conclusion

For web-based security, SSL/TLS is the better choice because it directly protects application-layer communication. For network-layer and VPN protection, IPsec is more appropriate. Therefore, the best enterprise solution can use both: TLS for web applications and IPsec for network-level protection.

2. End-to-End Encryption vs Gateway-Based Encryption

Scenario

An enterprise needs a secure email system for communication between employees, customers and external organizations.

Comparison

Feature	End-to-End Encryption	Gateway-Based Encryption
Encryption point	Sender to recipient	Email gateway
Decryption	Recipient's device	Gateway/server
Key control	Users/endpoints	Organization/gateway
Security	Very high against intermediaries	Depends on gateway security
Enterprise management	More difficult	Easier
Central monitoring	Limited	Easy
Data loss prevention	More difficult	Easier
Compliance management	More complex	Easier
Scalability	Requires key management	Easier for large organizations
Example	PGP/S/MIME	Secure email gateway

End-to-End Encryption

In end-to-end encryption, the message is encrypted by the sender and remains encrypted until it reaches the intended recipient.

Advantages:

1. Protects email contents from intermediaries.
2. Provides strong confidentiality.
3. Reduces exposure if intermediate servers are compromised.
4. Provides better privacy.

Disadvantages:

1. Key management can be difficult.
2. Password/key recovery can become complicated.
3. Enterprise monitoring is more difficult.

Gateway-Based Encryption

In gateway-based encryption, an organization's email gateway encrypts and decrypts messages.

Advantages:

1. Centralized security management.
2. Easier monitoring and auditing.
3. Easier integration with spam and malware filtering.
4. Suitable for enterprise policies.

Disadvantages:

1. Gateway becomes a critical security point.
2. Compromise of the gateway may expose messages.
3. Does not provide the same endpoint-to-endpoint protection as true E2E encryption.

Conclusion

For maximum confidentiality, end-to-end encryption is stronger. However, for a large enterprise requiring centralized monitoring, compliance, filtering and policy enforcement, gateway-based encryption is generally more practical.

A hybrid approach can provide the best balance: gateway security controls combined with end-to-end encryption for highly sensitive communications.

3. HMAC vs Digital Signature

Scenario

A company sends an important transaction from a client to a server. The system must ensure that the message has not been modified and that the sender can be authenticated.

Comparison

Feature	HMAC	Digital Signature
Cryptography	Symmetric	Asymmetric
Keys	Shared secret key	Public/private key pair
Integrity	Yes	Yes
Authentication	Yes	Yes
Non-repudiation	No	Yes
Speed	Faster	Slower
Key distribution	Difficult for many users	Easier
Computational cost	Low	Higher
Suitable for	Internal systems	Public/large-scale communication
Example	HMAC-SHA-256	RSA/ECDSA signature

HMAC Working

HMAC combines a secret key with a hash function.

For example:

HMAC-SHA-256(Message, Secret Key)

The sender generates the HMAC and sends it with the message. The receiver calculates the HMAC again using the shared secret key.

If both values match:

Message is authentic + Message integrity is maintained

Digital Signature Working

The sender generates a hash of the message and signs the hash using a private key.

The receiver uses the sender's public key to verify the signature.

Therefore:

Private Key → Sign → Digital Signature

Public Key → Verify → Authenticity

Critical Analysis

HMAC is faster and computationally efficient, making it suitable for high-speed internal communication. However, both parties must possess the same secret key. Digital signatures are computationally more expensive but provide a major additional benefit: non-repudiation. The sender cannot easily deny having signed the message because the signature is generated using the sender's private key.

Conclusion

If the main requirements are speed, authentication and integrity, HMAC is preferable. If the requirements include authentication, integrity and non-repudiation, a digital signature is the better choice.

Therefore, for important financial or legal transactions, digital signatures provide stronger security.

4. Supervised vs Unsupervised Learning for Spam Detection

Scenario

An email service receives thousands of emails every day. The system must automatically identify spam and legitimate messages.

Comparison

Feature	Supervised Learning	Unsupervised Learning
Training data	Requires labeled data	Does not require labeled data
Spam examples	Known during training	Not explicitly provided
Accuracy	Generally high with good data	Usually lower for precise classification
Training	Requires preparation of labeled dataset	Uses clustering/pattern detection
Adaptability	Needs retraining	Can discover new patterns
Classification	Direct	Indirect
Example algorithms	Naive Bayes, SVM, Decision Tree	K-Means, clustering
Maintenance	Requires labeled updates	Can operate with less labeling

Feature	Supervised Learning	Unsupervised Learning
Best use	Known spam patterns	Unknown/new patterns

Supervised Learning

A supervised spam detector is trained using examples such as:

- Spam: "Congratulations! You won a prize."
- Spam: "Click here to claim your reward."
- Not Spam: "Meeting scheduled for tomorrow."
- Not Spam: "Please submit the assignment."

The algorithm learns patterns from these labeled examples.

Advantages

1. High classification accuracy.
2. Directly predicts spam/not spam.
3. Easy to evaluate using known labels.
4. Suitable for large email services.

Disadvantages

1. Requires large labeled datasets.
2. New spam patterns may require retraining.
3. Incorrect labels can reduce accuracy.

Unsupervised Learning

Unsupervised learning identifies groups or patterns without predefined labels.

For example, emails with similar characteristics can be grouped into clusters. A cluster containing suspicious characteristics may be identified as potential spam.

Advantages

1. Does not require manually labeled data.
2. Can discover previously unknown patterns.
3. Useful for detecting emerging spam behavior.

Disadvantages

1. Clusters do not automatically indicate "spam" or "not spam."
2. Accuracy can be difficult to measure.
3. Human interpretation may be required.

Conclusion

For real-world email filtering, supervised learning is generally more effective for primary spam classification because it learns directly from labeled spam and legitimate emails.

However, unsupervised learning can complement it by detecting new or unusual spam patterns.

Recommended System

Supervised classifier + Unsupervised anomaly detection

This hybrid approach can provide better adaptability and detection capability.

5. AES vs RSA for Secure Communication

Scenario

Consider an online banking system where confidential customer information must be transmitted securely. The system requires both strong encryption and efficient communication.

Comparison

Feature	AES	RSA
Type	Symmetric encryption	Asymmetric encryption
Keys	Single shared key	Public and private keys
Speed	Very fast	Slower
Encryption of large data	Excellent	Not efficient
Key distribution	Difficult	Easier
Security purpose	Confidentiality	Encryption/key exchange/signatures
Computational cost	Low	High
Typical key sizes	128/192/256 bits	Commonly 2048 bits or higher
Suitable for	Bulk data	Small data/keys
Example use	File and network encryption	Key exchange, digital signatures

AES

AES uses the same secret key for encryption and decryption.

Plaintext → AES + Secret Key → Ciphertext

The receiver uses the same key to decrypt the ciphertext.

AES is highly efficient and is suitable for encrypting large amounts of information such as:

- Banking transactions
- Files
- Databases
- Network traffic
- Cloud data

RSA

RSA uses two keys:

Public Key → Encryption/Verification

Private Key → Decryption/Signing

RSA is slower than AES and is therefore not normally used to encrypt large amounts of data directly.

It is useful for:

- Secure key exchange
- Digital signatures
- Authentication
- Establishing secure sessions

Critical Analysis

AES provides excellent efficiency because it uses symmetric cryptography. Its main challenge is securely sharing the secret key.

RSA solves the key-distribution problem through public-key cryptography but requires considerably more computational resources.

Best Solution: Hybrid Encryption

A secure communication system can combine both techniques:

Step 1: Generate a random AES session key.

Step 2: Encrypt the actual data using AES.

Step 3: Encrypt the AES session key using the recipient's RSA public key.

Step 4: Send the encrypted data and encrypted AES key.

Step 5: Receiver uses the RSA private key to recover the AES key.

Step 6: AES key decrypts the actual data.

Conclusion

AES is preferred for efficient bulk-data encryption, while RSA is useful for secure key exchange and digital signatures.

Therefore, combining AES and RSA provides a better balance of confidentiality, security and efficiency.

Overall Comparative Conclusion

The five comparisons demonstrate that cryptographic and security mechanisms should be selected according to the security requirement rather than using one technique for every situation.

Security Requirement	Recommended Technique
Web security	SSL/TLS
Network/VPN security	IPsec
Maximum email confidentiality	End-to-End Encryption
Enterprise email management	Gateway-Based Encryption
Fast authentication & integrity	HMAC
Authentication + non-repudiation	Digital Signature
Primary spam classification	Supervised Learning
Detection of unknown patterns	Unsupervised Learning
Fast bulk encryption	AES
Key exchange/signatures	RSA

Thus, effective cybersecurity generally depends on combining complementary technologies, such as TLS with IPsec, HMAC with secure key management, supervised and unsupervised spam detection, and AES with RSA.